

Part I. Calculus I competency.

1. (5 pts) Find the equation of the tangent line to the graph of the function $f(x) = 3x - 4\sqrt{x}$ at $x = 4$.
2. (5 pts) Find the derivative of $a(x) = \frac{7}{x^3} - 10x^{3/4}$
3. (5 pts) Find the derivative of $b(x) = \frac{3x+1}{2-5x}$
4. (5 pts) Find the derivative of $c(x) = e^{-x} \sin 5x$
5. (5 pts) Find the derivative of $e(x) = x \ln x - x^2$
6. (5 pts) Find the derivative of $f(x) = (x^3 + 1)^2$
7. (5 pts) Find the indefinite integral $\int (\sqrt[3]{x} + x^3) dx$
8. (5 pts) Find the definite integral of $\int_0^{\pi/2} (1 - \cos x) dx$
9. (5 pts) Find the indefinite integral $\int (4x - 1)^3 dx$
10. (5 pts) Find the indefinite integral $\int \frac{\ln x}{3x} dx$

Part II. Calculus II Competency.

11. (5 pts) Find the area bounded between the graphs of $y = x^4$ and $y = x$.
12. (5 pts) Find the indefinite integral $\int x \sin x dx$
13. (5 pts) Find the partial fraction decomposition for the function $f(x) = \frac{2x+1}{x^2-x}$
14. (5 pts) Find the indefinite integral $\int \frac{2x-1}{(x-x^2)^4} dx$
15. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} \frac{1}{k^2 + k + 1}$
16. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} k^2 \left(\frac{3}{4}\right)^k$
17. (5 pts) Find the first four non-zero terms of the MacLaurin series of $f(x) = \cos(x^2)$

Part III. Calculus III competency.

18. (5 pts) Find f_x, f_y for $f(x, y) = xye^{x+y^2}$
19. (5 pts) Find f_{xx}, f_{yx} for $f(x, y) = xye^{x+y^2}$
20. (5 pts) Find f_x, f_y, f_z for $f(x, y, z) = \frac{xz - y^2}{z^4}$